

Flutter Assignment 4 (Q1–Q5) – Code

1) Counter App using Provider

counter_provider.dart

```
import 'package:flutter/material.dart';

class CounterProvider extends ChangeNotifier {
  int count = 0;

  void increment() {
    count++;
    notifyListeners();
  }

  void decrement() {
    count--;
    notifyListeners();
  }
}
```

main.dart

```
import 'package:flutter/material.dart';
import 'package:provider/provider.dart';
import 'counter_provider.dart';

void main() {
  runApp(
    ChangeNotifierProvider(
      create: (_) => CounterProvider(),
      child: MyApp(),
    ),
  );
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(home: Home());
  }
}

class Home extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    var counter = Provider.of<CounterProvider>(context);

    return Scaffold(
      appBar: AppBar(title: Text("Counter")),
      body: Center(child: Text("${counter.count}")),
      floatingActionButton: Column(
        mainAxisAlignment: MainAxisAlignment.end,
        children: [
          FloatingActionButton(onPressed: counter.increment, child: Icon(Icons.add)),
          SizedBox(height: 10),
        ],
      ),
    );
  }
}
```

```

        FloatingActionButton(onPressed: counter.decrement, child: Icon(Icons.remove)),
      ],
    ),
  );
}
}

```

2) Dynamic Text (Multi Screen)

text_provider.dart

```

import 'package:flutter/material.dart';

class TextProvider extends ChangeNotifier {
  String text = "Guest";

  void changeText() {
    text = "User ${DateTime.now().second}";
    notifyListeners();
  }
}

```

main.dart

```

runApp(
  ChangeNotifierProvider(
    create: (_) => TextProvider(),
    child: MaterialApp(home: Home()),
  ),
);

```

home.dart

```

class Home extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    var data = Provider.of<TextProvider>(context);

    return Scaffold(
      body: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          Text(data.text),
          ElevatedButton(onPressed: data.changeText, child: Text("Change")),
          ElevatedButton(
            onPressed: () => Navigator.push(
              context, MaterialPageRoute(builder: (_) => Profile()),
              child: Text("Next"),
            ),
          ),
        ],
      ),
    );
  }
}

```

profile.dart

```

class Profile extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    var data = Provider.of<TextProvider>(context);
    return Scaffold(body: Center(child: Text(data.text)));
  }
}

```

3) Light / Dark Theme

theme_provider.dart

```

import 'package:flutter/material.dart';

class ThemeProvider extends ChangeNotifier {
  bool isDark = false;

  void toggle() {
    isDark = !isDark;
    notifyListeners();
  }
}

```

main.dart

```

runApp(
  ChangeNotifierProvider(
    create: (_) => ThemeProvider(),
    child: MyApp(),
  ),
);

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    var theme = Provider.of<ThemeProvider>(context);

    return MaterialApp(
      theme: theme.isDark ? ThemeData.dark() : ThemeData.light(),
      home: Home(),
    );
  }
}

```

home.dart

```

class Home extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    var theme = Provider.of<ThemeProvider>(context);

    return Scaffold(
      body: Center(
        child: Switch(
          value: theme.isDark,
          onChanged: (_) => theme.toggle(),

```

```

    ),
  ),
);
}
}

```

4) Notes App (SQLite)

db_helper.dart

```

import 'package:sqflite/sqflite.dart';
import 'package:path/path.dart';

class DBHelper {
  static Future<Database> db() async {
    return openDatabase(
      join(await getDatabasesPath(), 'notes.db'),
      onCreate: (db, v) {
        return db.execute(
          "CREATE TABLE notes(id INTEGER PRIMARY KEY, title TEXT, description TEXT)",
        );
      },
      version: 1,
    );
  }

  static Future<void> insert(String t, String d) async {
    final database = await db();
    await database.insert("notes", {"title": t, "description": d});
  }

  static Future<List<Map<String, dynamic>>> getAll() async {
    final database = await db();
    return database.query("notes");
  }

  static Future<void> delete(int id) async {
    final database = await db();
    await database.delete("notes", where: "id=?", whereArgs: [id]);
  }
}

```

5) Login (SharedPreferences)

main.dart

```

import 'package:flutter/material.dart';
import 'package:shared_preferences/shared_preferences.dart';

void main() => runApp(MyApp());

class MyApp extends StatelessWidget {
  Future<bool> check() async {
    final p = await SharedPreferences.getInstance();
    return p.getBool("login") ?? false;
  }
}

```

```

@override
Widget build(BuildContext context) {
  return MaterialApp(
    home: FutureBuilder(
      future: check(),
      builder: (c, s) {
        if (!s.hasData) return CircularProgressIndicator();
        return s.data! ? Home() : Login();
      },
    ),
  );
}
}

```

login.dart

```

class Login extends StatelessWidget {
  void login(BuildContext c) async {
    final p = await SharedPreferences.getInstance();
    await p.setBool("login", true);
    Navigator.pushReplacement(c, MaterialPageRoute(builder: (_) => Home()));
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Center(
        child: ElevatedButton(
          onPressed: () => login(context),
          child: Text("Login"),
        ),
      ),
    );
  }
}

```

home.dart

```

class Home extends StatelessWidget {
  void logout(BuildContext c) async {
    final p = await SharedPreferences.getInstance();
    await p.clear();
    Navigator.pushReplacement(c, MaterialPageRoute(builder: (_) => Login()));
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: Center(
        child: ElevatedButton(
          onPressed: () => logout(context),
          child: Text("Logout"),
        ),
      ),
    );
  }
}

```